



Break Point of an Hypothesis Set

- Given an hypothesis set $\mathcal{H}$
- A hypothesis set $\mathcal{H}$ **shatters N points** $\iff m_{\mathcal{H}}(N) = 2^N$
 - There is a position of N points and a class assignment that you can classify using $h \in \mathcal{H}$
 - It does not mean all sets of N points can be classified in any way
- k is a **break point** for $\mathcal{H}$ $\iff m_{\mathcal{H}}(k) < 2^k$
 - I.e., no data set of size k can be shattered by $\mathcal{H}$
 - E.g.,
 - For 2D perceptron: a break point is 4
 - For positive rays: a break point is 2
 - For positive intervals: a break point is 3
 - For convex set on a plane: there is no break point

Break Point for an Hypothesis Set and Learning

- If there is a break point for a hypothesis set $\mathcal{H}$, it can be shown that:
 - $m_{\mathcal{H}}(N)$ is polynomial in N
 - Instead of Hoeffding's inequality for learning

$$\Pr(|E_{in}(g) - E_{out}(g)| > \epsilon) \leq 2Me^{-2\epsilon^2 N}$$

you can use the Vapnik-Chervonenkis inequality:

$$\Pr(\text{bad generalization}) \leq 4m_{\mathcal{H}}(2N)e^{-\frac{1}{8}\epsilon^2 N}$$

- Since $m_{\mathcal{H}}(N)$ is polynomial in N , it will be dominated by the negative exponential, given enough examples
- You can have a generalization bound: machine learning works!
- A hypothesis set can be characterized from the learning point of view by the **existence and value of a break point**

- *The VC Dimension*
- Overfitting

VC Dimension of an Hypothesis Set

- The **VC dimension of a hypothesis set** $\mathcal{H}$, denoted as $d_{VC}(\mathcal{H})$, is defined as the largest value of N for which $m_{\mathcal{H}}(N) = 2^N$
 - I.e., the VC dimension is the most points $\mathcal{H}$ can shatter
- **Properties** of the VC dimension: if $d_{VC}(\mathcal{H}) = N$ then
 - Exists a constellation of N points that can be shattered by $\mathcal{H}$
 - Not all sets of N points can be shattered
 - If N points were placed randomly, they could not be necessarily shattered
 - $\mathcal{H}$ can *shatter* N points for any $N \leq d_{VC}(\mathcal{H})$
 - The *smallest break point* is $d_{VC} - 1$
 - The *growth function* in terms of the VC dimension is $m_{\mathcal{H}} \leq \sum_{i=0}^{d_{VC}} \binom{N}{i}$
 - The VC dimension is the *order of the polynomial bounding* $m_{\mathcal{H}}$

VC Dimension: Interpretation

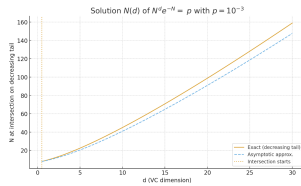
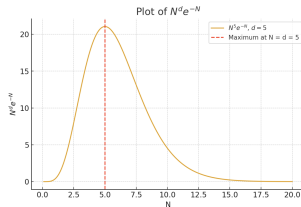
- The VC dimension **measures the complexity** of a hypothesis set in terms of **effective parameters**
- E.g.,
 - A perceptron in a d -dimensional space has $d_{VC} = d + 1$
 - In fact d_{VC} is the number of perceptron parameters!
 - E.g., for a 2D perceptron ($d = 2$), the break point is 2, so $d_{VC} = 3$
- The VC dimension considers the model as a black box in order to estimate effective parameters
 - How many points N a model can shatter, not the number of parameters
- Not all parameters contribute to degrees of freedom
 - E.g., combining N 1D perceptrons gives $2N$ parameters, but the effective degrees of freedom remain 2
- A complex hypothesis $\mathcal{H}$:
 - Has more parameters (higher VC dimension d_{VC})
 - Requires more examples for training

VC Generalization Bounds

- How many data points are needed to obtain $\Pr(|E_{in} - E_{out}| > \varepsilon) \leq \delta$?
- The VC inequality states

$$\Pr(\text{bad generalization}) \leq 4m_{\mathcal{H}}(2N)e^{-\frac{1}{8}\varepsilon^2 N}$$

- $N^d e^{-N}$ abstracts the upper bound term
 - Plot $N^d e^{-N}$ vs. N : Power dominates for small N , exponential for large N and brings it to 0
 - Vary d (VC dimension) function peaks for larger N , then approaches the region of interest $N^d e^{-N} \leq N^{-1}$ with a probability as a function of d
 - Examples N needed are proportional to d
 - Rule of thumb: $N \geq 10d_{VC}$ for generalization



VC Generalization Bounds

- The VC inequality

$$\Pr(|E_{in} - E_{out}| > \varepsilon) \leq 4m_{\mathcal{H}}(2N)e^{-\frac{1}{8}\varepsilon^2 N}$$

can be used in several ways to relate ε , δ , and N , e.g.,

- Examples
 - “Given $\varepsilon = 1\%$ error, how many examples N are needed to get $\delta = 0.05$?”
 - “Given N examples, what’s the probability of an error larger than ε ?”
- You can equate δ to $4m_{\mathcal{H}}(2N)e^{\frac{1}{8}\varepsilon^2 N}$ and solve for ε , getting

$$\Omega(N, \mathcal{H}, \delta) = \sqrt{\frac{8}{N} \ln \frac{4m_{\mathcal{H}}(2N)}{\delta}}$$

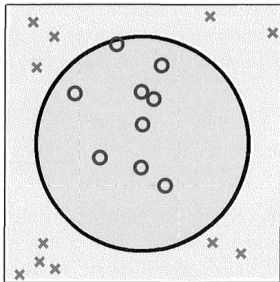
- Then you can say $|E_{out} - E_{in}| \leq \Omega(N, \mathcal{H}, \delta)$ with probability $\geq 1 - \delta$
 - The generalization bounds are then: $\Pr(E_{out} \leq E_{in} + \Omega) \geq 1 - \delta$

How to Void the VC Analysis Guarantee

- Consider the case where data is genuinely non-linear
 - E.g., “o” points in the center and “x” in the corners
- Transform to high-dimensional $\mathcal{Z}$ with:

$$\Phi : \underline{x} = (x_0, \dots, x_d) \rightarrow \underline{z} = (z_0, \dots, z_{\tilde{d}})$$

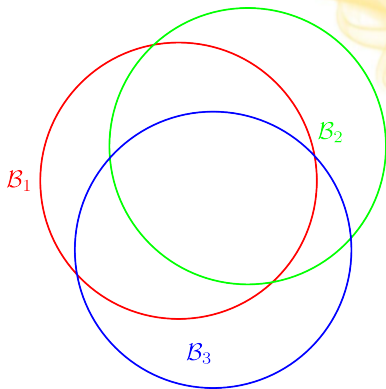
- $d_{VC} \leq \tilde{d} + 1$; smaller $\tilde{d}$ improves generalization
 - Use $\underline{z} = (1, x_1, x_2, x_1 x_2, x_1^2, x_2^2)$
 - Why not $\underline{z} = (1, x_1^2, x_2^2)$?
 - Why not $\underline{z} = (1, x_1^2 + x_2^2)$?
 - Why not $\underline{z} = (x_1^2 + x_2^2 - 0.6)$?
- Some model coefficients were zero and discarded, leaving machine learning the rest
 - VC analysis is a warranty, forfeited if data is examined before model selection (data snooping)
 - From VC analysis, complexity is that of the initial hypothesis set



- The VC Dimension
- *Overfitting*

Overfitting: Definition

- Overfitting occurs when the model fits the data more than what is warranted
- In practice, we surpass the point where E_{out} is minimal (optimal fit)
 - Model complexity is too high for the data/noise
 - Noise in the training set is mistaken for signal
- Fitting the noise instead than signal is worse than useless, it is harmful
 - The model infers a pattern in-sample that, when extrapolated out-of-sample, deviates from the target function $\Rightarrow$ poor generalization



Optimal Fit

- “Optimal fit” is the opposite of overfitting
 - Train a model with the proper complexity for the data
- The optimal fit:
 - Implies that E_{out} is minimal
 - Does not imply that generalization error $E_{out} - E_{in}$ is minimal (e.g., no training at all implies generalization error equal to 0)
- The generalization error is the additional error $E_{out} - E_{in}$ we see when we go from in-sample to out-of-sample

Overfitting: Diamond Price Example

- Predict diamond price as a function of carat size (regression problem)
- True relationship:

$$\text{price} \sim (\text{carat size})^2 + \varepsilon$$

where:

- Square function: price increases more with rarity
- Noise: e.g., market noise, missing features

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- **Fit with:**

- Line $\rightarrow$ underfit
 - High bias (large error)
 - Low variance (stable model)
- Polynomial of degree 2 $\rightarrow$ right fit
- Polynomial of degree 10 $\rightarrow$ overfit (wiggly curve)
 - Low bias
 - High variance (many degrees of freedom) :::: :::: {column width=40%}

